

MTH2021 Mathematical Methods 2

Homework set 5

1. If $f(z)$ has a pole of order m at the point z_0 , then the Laurent series of $f(z)$ around z_0 reads

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots$$

Show that the residue of $f(z)$ at the pole, namely the coefficient of $(z - z_0)^{-1}$, a_{-1} , is given by

$$a_{-1} = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

2. Find the residues of each of the following functions at its poles:

(a)

$$f(z) = \frac{5z - 2}{z(z - 1)}$$

(b)

$$f(z) = \frac{z^2 - 2z + 3}{(z - 2)^2}$$

(c)

$$f(z) = \frac{\sinh z}{z^4}$$

3. (a) Show that for $a > 0$

$$\int_0^\infty \frac{x^2}{(a^2 + x^2)^3} dx = \frac{\pi}{16a^3}$$

(b) Evaluate the integral

$$\int_{-\infty}^\infty \frac{x^2}{1 + x^4} dx$$

4. Consider the integral

$$I = \int_0^{2\pi} \frac{d\theta}{5 + 4 \cos \theta}$$

By changing the integration variable from θ to the complex variable $z = \exp(i\theta)$, show that

$$I = \frac{1}{i} \oint_C \frac{dz}{(2z + 1)(z + 2)}$$

where C is the closed contour $|z| = 1$.

Using Cauchy's residue theorem, show that $I = \frac{2\pi}{3}$.

5. Using the residue theorem, show that

$$\int_{-\infty}^\infty \frac{\cos x}{(1 + x^2)^2} dx = \frac{\pi}{e}$$
